

I had written this report in Markdown, accessible as the .md file in my zip submission, and I took the output and pasted it here (auto-formats with title / sections / table rendering), rendered below.

PSET 2 Final Report: SQL-Based Business Analytics for NJ Teacher Salaries

Overview

For this assignment, I worked through four business scenarios using the NJ Teachers Salaries dataset. The idea was to write SQL queries, pull out meaningful patterns from the data, and then turn those patterns into actionable recommendations. Each scenario addresses a different education policy question, and I tried to keep my analysis grounded in what the numbers actually reveal rather than making assumptions.

Scenario 1: Budget Optimization in School Districts

Problem Statement: A school district is dealing with budget constraints and needs to figure out whether its salary allocations make sense. Before recommending any changes, I wanted to first understand the current landscape - which districts pay the most, which pay the least, and how salary grows with experience.

Task 1: Salary Distribution Analysis

Top 5 Districts by Average Salary:

District	Average Salary
Northern Valley Regional	\$101,840.40
Passaic County Vocational	\$95,298.63
Pascack Valley Regional	\$95,191.84
Palisades Park	\$93,754.00
Hackensack City	\$92,813.21

Bottom 5 Districts by Average Salary:

District	Average Salary
West Cape May Boro	\$43,184.27
Classical Academy Charter School	\$45,000.00
Hope Community Cs	\$45,762.33
Hudson Arts And Science Charter School	\$46,439.29
The Kingdom Charter School Of Leadership	\$46,807.83

Task 2: Experience vs. Salary Analysis

Average Salary by Years of Experience (First 10 Years):

Year	Average Salary
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0	\$55,158.68
1	\$57,323.85
2	\$57,567.87
3	\$58,297.50
4	\$59,165.91
5	\$60,408.58
6	\$61,454.77
7	\$62,983.57
8	\$65,514.38
9	\$67,338.71

Task 3: Recommendations

1. Right away, the gap between the highest and lowest paying districts jumped out at me - nearly \$58,000, which is more than double. That kind of disparity is hard to ignore and probably deserves a closer look at how funding gets distributed across the state.
 2. What also caught my attention is that 4 of the 5 lowest-paying districts are charter schools. That feels like more than a coincidence, and it suggests charter school compensation might need its own dedicated policy conversation.
 3. On the experience side, salary growth is pretty steady - around \$1,200 per year on average. One way to address the district-level disparity could be standardizing salary bands tied to experience, so teachers aren't penalized just because of where they happen to work.
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Scenario 2: Retention of High-Performing Teachers

Problem Statement: Passaic County is worried about losing experienced teachers to neighboring counties or out-of-state jobs. The concern makes sense on the surface, but I wanted to check whether the data actually supports it. Is Passaic really falling behind, or is this more perception than reality?

Task 1: Passaic County Salary vs Experience

Passaic Avg Salary for Years 10-15:

Year	Average Salary
10	\$68,490.36
11	\$73,041.86
12	\$76,403.78
13	\$78,529.07
14	\$80,081.32
15	\$84,373.50

Task 2: Passaic Retention Strategy (>10 Years Experience)

Metric	Value
Passaic Avg Salary (>10 Years)	\$90,709.14

Passaic Teacher Count (>10 Years)	8,238
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Task 3: Cross-County Comparison (>10 Years Experience)

Average Salary for Experienced Teachers (>10 Years):

County	Average Salary
Bergen	\$93,020.49
Passaic	\$90,709.14
Essex	\$89,280.30
Morris	\$86,276.51
Sussex	\$86,153.71

Task 4: Retention Strategy Recommendations

1. Honestly, Passaic is doing better than I expected. It ranks 2nd out of the 5 neighboring counties, and the gap with Bergen at the top is only about \$2,300. That's close enough that a few targeted incentives could close it without needing a major budget overhaul.
 2. Passaic also has a solid base to work with - 8,238 teachers with over 10 years of experience. And the salary growth from year 10 (\$68k) to year 15 (\$84k) is healthy, roughly \$3,200 per year. That progression probably helps explain why retention isn't as bad as feared.
 3. Going forward, I'd recommend Passaic focus on protecting its lead over Essex, Morris, and Sussex while looking for small ways to close the gap with Bergen. Milestone bonuses at year 10 or 15 could be enough to make the difference.
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Scenario 3: Cost-Benefit Analysis of Certification Programs

Problem Statement: The state education board wants to know if funding certification programs is worth it. The logic seems straightforward - if certified teachers earn more and stay longer, then helping teachers get certified should pay for itself through reduced turnover. But I wanted to see what the data actually shows before jumping to conclusions.

Task 1: Salary Difference by Certification

Certification Status	Average Salary
Standard	\$75,824.50
Non-Standard	\$59,497.05

Task 2: Average Experience by Certification

Certification Status	Average Experience
Standard	13.2 years
Non-Standard	3.7 years

Task 3 & 4: Cost-Benefit Recommendations

1. There's definitely a salary gap - about \$16,000 between standard and non-standard certification. But here's the thing: standard certified teachers also have nearly a decade more experience on average (13.2 years vs 3.7). So it's hard to know how much of that salary difference comes from the certification itself versus just having been around longer.

- 2. That 9.5-year experience gap tells me that most non-standard teachers are probably early in their careers and still working toward full certification. If the state can help them get there faster, it could improve retention and save money on turnover in the long run.
- 3. Taking all of this together, I think investing in certification programs makes sense. Even if we can't perfectly isolate the effect of certification, the data suggests certified teachers stick around longer. And given how expensive turnover is - recruiting, training, onboarding - the math probably works out in favor of funding these programs.

Scenario 4: Optimal Salary Structures for New Teachers

Problem Statement: The Department of Education wants to attract more recent graduates into teaching and is thinking about how to structure early-career salaries. The real question I wanted to answer is: where exactly are we losing new teachers, and would paying them more at the right moments actually help?

Task 1: Starting Salary Analysis (0-5 Years)

Metric	Value
Average Salary for New Teachers (0-5 Years)	\$57,904.97

Task 2: Salary Progression (Year 0 to 5)

Year	Average Salary
0	\$55,158.68
1	\$57,323.85

2	\$57,567.87
3	\$58,297.50
4	\$59,165.91
5	\$60,408.58

Task 3: Retention Factors (Teacher Count Drop-off)

Year	Teacher Count
0	12,140
1	16,328
2	13,612
3	13,364
4	11,858
5	10,872
6	8,462
7	9,012
8	10,428
9	11,156

10	11,706
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Task 4: Salary Structure Recommendations

1. The data surprised me here. I expected the biggest drop-off to be in the first couple years, but it's actually between year 5 and year 6 - teacher count falls from 10,872 to 8,462, a 22% decline in a single year. So if we're going to introduce a retention bonus or milestone incentive, year 5 seems like the place to do it.
2. That said, there's also something going on early. Salary growth from year 1 to year 2 is almost flat - just \$244. Meanwhile, we lose nearly 2,700 teachers during that same window. Bumping the year 2 raise might help retain people through that early transition period.
3. One more thing I noticed: after year 6, the numbers actually start climbing back up through year 10. It seems like teachers who make it past that 6-year mark tend to stick around. That reinforces the idea that early-career retention is where the effort needs to go - if you can keep them through year 6, you've probably got them for the long haul.

Conclusion

Looking back at all four scenarios, a few things stand out to me:

- The district salary gap is striking. Nearly \$58,000 separates the top from the bottom, and charter schools cluster at the low end. That's not going to fix itself without some deliberate policy intervention.
- Certification and experience are hard to untangle. Certified teachers earn more, but they're also much more experienced on average. I can't say for certain how much certification alone matters, but helping teachers get certified faster still seems like a reasonable bet given the retention benefits.
- The retention cliff is at year 5-6, not year 1-2. That was the biggest surprise for me. If the goal is to keep teachers in the profession, that's the window to target with incentives.
- Passaic is actually in decent shape. It's 2nd among its neighbors for experienced teacher salaries, only \$2,300 behind Bergen. A few small, targeted moves could close that gap without blowing up the budget.

At the end of the day, the data points toward a few high-leverage interventions: address charter school pay, support paths to certification, and focus retention incentives around the year 5-6 mark. None of these are dramatic overhauls - they're targeted adjustments based on where the numbers suggest the problems actually are.